

Multi-agent distributed formation control using local information measurements

Jianqing Li¹, Liaofei Chen^{2*} & Chaoyong Li³

¹Space Information Research Institute and Zhejiang Key Laboratory of Space Information Sensing and Transmission, Hangzhou Dianzi University, Hangzhou 310018, China;

²College of Communication Engineering, Hangzhou Dianzi University, Hangzhou 310018, China;

³College of Electrical Engineering, Zhejiang University, Hangzhou 310027, China

Received January 11, 2016; accepted April 6, 2016; published online January 1, 2016

This paper investigates formation control in multi-agent systems to adjust deformation and scaling for obstacle avoidance using local information. A well-known issue with angle-based formation control is the existence of size ambiguity. Introducing additional controlled variables can potentially mitigate this problem. Therefore, unlike purely angle-based formation control methods that can only adjust deformation, our approach can also control scaling by introducing distance measurements. Specifically, all agents use their local coordinate systems and sense neighboring agents to obtain angle measurements, with only one agent measuring the distance to its neighbor. In the proposed formation control scheme, angle measurements are used to control the desired shape, while distance measurements regulate the formation scaling. Different formation control strategies for triangular and polygonal configurations are discussed, utilizing angle rigidity to ensure the stability of the final shape in polygonal configurations with only one distance measurement used for formation scaling. Additionally, by relaxing the condition of angular rigidity and utilizing the concept of formation rigidity, we propose a formation determination method to construct a rigid square configuration, in which a pair of distances measured by one agent are used to control the formation's scaling. Finally, using a simplified model of a quadrotor Unmanned Aerial Vehicle (QUAV), simulation results validate the effectiveness of the proposed formation control scheme through several examples.

angle measurements, distance measurements, formation control, rigid formation, multi-agent systems

Citation: Li J Q, Chen L F, Li C Y. Multi-agent distributed formation control using local information measurements. *Sci China Tech Sci*, 2017, 60, <https://doi.org/10.1007/s11432-016-0037-0>

1 Introduction

In recent years, collaborative control of multi-agent systems has attracted widespread attention, with formation control issues receiving extensive research due to their significance across multiple application domains [1, 2]. Examples include UAV search and rescue [3], collaborative transport of ground robots [4], and satellite formation flying [5].

A key focus is addressing which variables need to be

sensed and actively controlled to achieve desired formations in multi-agent systems, as mentioned in [6]. Sensors used for formation control typically include the Global Positioning System (GPS), radar, and cameras, which can measure positions, inter-agent distances, and bearings, respectively. The existing formation control methods are categorized based on the sensors and control variables used, including position-based, displacement-based (consensus-based) [7, 8], direction-based, distance-based, and angle-based approaches. However, a disadvantage of position-based forma-

*Corresponding author (email: liaofei@hdu.edu.cn)

tion control is that it typically requires a global coordinate system shared among the agents [9]. In displacement-based methods, agents measure the relative positions of their neighbors in a common reference coordinate system, necessitating that all agents have a shared sense of direction [10, 11]. This implies that the local coordinate frames of all agents need to be aligned [12]. Achieving perfect alignment of all agents' coordinate systems is challenging due to the presence of sensor measurement noise and interference. When there is a misalignment in the agents' coordinate frames, the formation shape specified by relative positions may be distorted and exhibit non-zero translational velocity [13, 14]. In [15], the theory of bearing rigidity was established, but it still requires agents to know a global reference frame. In contrast, distance/angle-based formation methods that rely solely on local information from multi-agent systems have become popular, even though they increase the complexity of their design and analysis.

Over the past few decades, distance rigidity has been extensively studied both as a mathematical topic in graph theory [16] and as an engineering problem in the application of multi-agent system formations. Distance rigidity [17] is defined based on the property that distances within a multi-point framework remain invariant under translational and rotational motions. Without requiring coordinate frame alignment, the desired formation shape described by distances was achieved using local relative position measurements in [18] by employing distance rigidity theory. Moreover, a theory based on bearing rigidity was proposed [15, 19]. Agents maintain their relative orientation with respect to their neighbors to ensure the stability of the formation. However, methods solely based on bearings become ambiguous due to the lack of measurement for formation size. To address this, a bearing-based approach within the leader-follower framework was introduced [20, 21], but it requires that the maneuvering strategies of two leaders be pre-planned. With the advancement of sensor technology, angle measurements can be easily obtained through omnidirectional radar or vision-based cameras, which are cheaper, more reliable, and more accessible than distance measurements [22, 23]. Since angle measurements are scalars and do not change with motion, including translation, rotation, and scaling [24, 25], the theory of angle rigidity was established in [26]. By assigning appropriate angle constraints to the agents, the desired formation shape can be achieved. However, a resulting issue is that the angles do not change with the scaling of the formation. When relying solely on angle information for control, the size of the formation becomes ambiguous when the multi-agent system stabilizes. To address this issue, a control algorithm for multi-agent systems was designed in [27], which determines

the formation shape through angle constraints while still using relative position measurements in the global coordinate system to scale the formation. Recently, the formation control method combining angle and distance has been studied in [28, 29], and a formation control scheme with directed graph distance measurements and angle constraints has been proposed. However, these methods impose complex directed graph distance constraints and angle measurements, increasing the difficulty of formation rigidity design. In many cases, it is necessary to manipulate the formation scale, such as to avoid obstacles [30], and the variation in pairwise distances between agents during scaling complicates the control algorithm.

Motivated by the work above, we propose a formation scaling control law based on angle and distance to address formation maintenance during obstacle avoidance. This approach requires each agent to measure only the angle information of its neighbors, with only one agent responsible for measuring distance information. Angle rigidity ensures the stability of the final formation shape, while distance control manages the scaling of the formation. The contributions of this paper are summarized as follows:

(1) Compared to [26], this addresses the issue of ambiguity in controlling the formation size using only angle information. In [26], when the formation system is stable, all angle constraints are satisfied. However, because angles do not change with scaling motions, the overall size of the formation is ambiguous and cannot be controlled arbitrarily. Additionally, to relax an angular rigidity constraint in [26], we discuss a method using a square formation case to control the scaling by weighting a pair of distances in two different directions.

(2) Different from the rigidity formation methods that combine angle and distance-directed graph in [28, 29], the scaling transformation of the formation in this paper is controlled by a single agent, eliminating the need to change a large number of desired distances between different agents and thereby simplifying the complexity of the control algorithm during the process of expansion and contraction motion.

(3) This work represents a preliminary step in addressing the problem of formation maintenance, deformation, and scaling control based on local information. Due to the difficulty in constructing a Lyapunov function, we have currently proven the local convergence of the closed-loop nonlinear formation under double integrator dynamics. In view of this, we provide sufficient conditions to ensure the stability of the double integrator formation, which offers valuable guidance for future research.

The remainder of this paper is organized as follows. Section II introduces the agent formation control problem. Sec-

tion III presents the main results with different configurations. Section IV shows the simulation results examining the performance of the proposed formation control law.

Notations: An undirected graph, denoted as $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, consists of a vertex set $\mathcal{V} = \{1, \dots, n\}$ and an edge set $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ with $m = |\mathcal{E}|$. The set of neighbors of vertex i is denoted as $\mathcal{N}_i \triangleq \{j \in \mathcal{V} : (i, j) \in \mathcal{E}\}$. An orientation of an undirected graph is the assignment of a direction to each edge. An oriented graph is an undirected graph together with an orientation.

2 Problem formulation

For the n mobile agents moving in the plane, we define the set of agents indexes as $\mathcal{V} = \{1, \dots, n\}$. The dynamics of each agent i can be described as

$$\begin{cases} \dot{p}_i = v_i, \\ \dot{v}_i = u_i, \end{cases} \quad \forall i \in \mathcal{V}, \quad (1)$$

where $p_i \in \mathbb{R}^2$ denotes the position of agent i in the fixed global coordinate system, $v_i \in \mathbb{R}^2$ is the velocity and $u_i \in \mathbb{R}^2$ is the control input.

The following assumptions are made for all agents to achieve the desired formation configuration.

Assumption 1 All agents are already on the same plane when the formation needs to deform and scale.

Assumption 2 Each agent is equipped with a sensor to measure LOS(Line of Sight).

Assumption 3 The initial positions of the agents are independent and non-collinear.

Assumption 1 ensures that the formation is in a 2-D plane. Assumption 2 requires each agent to measure the directional angle of its neighbor with respect to its local coordinate system. Assumption 3 guarantees that the angle bisector direction exists and is unique.

Under Assumptions 1-2, each agent i can measure the angle $\theta_{ij} \in [0, 2\pi)$ relative to $j \in \mathcal{N}_i$ in its local coordinate frame Σ_i , as shown in Fig. 1. The local coordinate system satisfies the right-hand frame. Then, the angle α_i in the x_i - y_i plane can be given by:

$$\alpha_i = \theta_{ij} - \theta_{ik}, \quad (2)$$

where $\alpha_i \in [0, 2\pi)$ is the interior angle of the triangle formed by j , i and k .

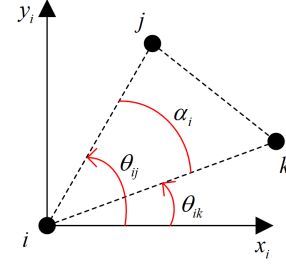


Figure 1 The angle measurements.

As shown in Fig. 1, the agent i can measure the LOS angle of its neighbors. Then, the interior angle α_i can be calculated by

$$\cos \alpha_i = z_{ij}^T z_{ik}, \quad (3)$$

where $z_{ij} = \frac{p_j - p_i}{\|p_j - p_i\|}$, which starts from i , points toward j .

Let $\alpha_i^* \in (0, \pi)$ be the desired steady-state angles of the formation as shown in Fig. 2(a). The angle-based formation problem becomes the realization of the desired angle constraint:

$$\lim_{t \rightarrow +\infty} (\alpha_i(t) - \alpha_i^*) = 0. \quad (4)$$

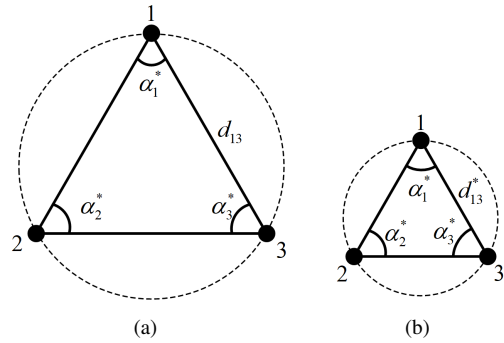


Figure 2 Uncertainty in formation scale based solely on angle control.

Definition 1 (Angle rigidity and Global angle rigidity) As mentioned in [26, Definition 2, Definition 3]

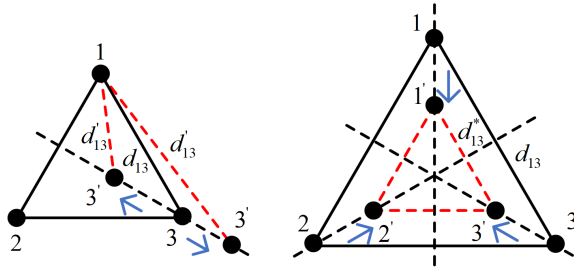
We give our description of the uniqueness of the framework based on graph theory.

Definition 2 (Similarity and Congruence) If two frameworks $\mathcal{G}_1 = (\mathcal{V}_1, \mathcal{E}_1, \mathcal{W}_1)$ and $\mathcal{G}_2 = (\mathcal{V}_2, \mathcal{E}_2, \mathcal{W}_2)$ satisfy the following conditions: (1) They have the same topological structure (i.e., there exists a graph isomorphism $f : \mathcal{V}_1 \rightarrow \mathcal{V}_2$, such that for every edge $(i, j) \in \mathcal{E}_1$, there is an edge $(f(i), f(j)) \in \mathcal{E}_2$). (2) For each pair of corresponding edges $e_{1i} \in \mathcal{E}_1$ and $e_{2i} \in \mathcal{E}_2$, their weights satisfy $w_{1i} = k \cdot w_{2i}$, where k is a fixed proportional constant; then it can be said that the

two frameworks are similar if $k \neq 1$, denoted as $\mathcal{G}_1 \sim \mathcal{G}_2$ and we say that two frameworks are congruent if $k = 1$, that is, $\mathcal{G}_1 \triangleq \mathcal{G}_2$.

However, a problem arises when relying solely on the constraints of (4); it cannot effectively control the formation scale. For instance, we observed that the angles used to constrain the configuration in Fig. 2(a) and Fig. 2(b) both satisfy (4), but they do not determine the distances between the agents. When the system is stable, it satisfies the angular rigidity as defined in Definition 1, but the actual framework $\mathcal{G}$ is not congruent with the desired framework $\mathcal{G}^*$. In other words, achieving the desired shape may result in the agents' formation scale being either too large or too small, that is, similarity as defined in Definition 2.

A triangle can be uniquely determined given its three angles and any one side. Therefore, for triangular formations, we can achieve the expansion and contraction of the formation by controlling the distance between any two agents while maintaining angle constraints. In this case, we specifically choose the distance d_{13} between agent #1 and agent #3.



(a) The scaling of the distance between agent #1 and agent #3. (b) Reduction in formation size.

Figure 3 The triangular formation motion strategy based on distance and angle.

As shown in Fig. 3(a), d_{13} will change while agent #1 and agent #2 remain stationary, and agent #3 moves along the straight line where the angle bisector of $\angle 132$ lies. Then, in Fig. 3(b), let us take the formation reduction as an example. agent #3 moves inward along the angle bisector, causing d_{13} to decrease. However, to ensure the constraints of the desired interior angles of the triangle, agent #1 and agent #2 will also move along the bisectors of their respective angles. As a result, the scale of the entire agent formation is reduced. To obtain d_{13} , we make the following assumption:

Assumption 4 Agent #3 can measure the distances between itself and the other agents. And the other agents do not need to measure distance information.

Problem 1 Consider three agents with double-integrator dynamics described by (1). The objective is to design a control law u_i for each agent i to facilitate the expansion and con-

traction of the formation while ensuring its stability. Then, extend the results to n agents. The stability of the entire agent system during formation maneuvers can be described as follows:

$$\begin{aligned} \lim_{t \rightarrow +\infty} (\alpha_i(t) - \alpha_i^*) &= 0, \\ \lim_{t \rightarrow +\infty} (v_i(t) - v_c^*) &= 0, \\ \lim_{t \rightarrow +\infty} (d_{ij}(t) - d_{ij}^*) &= 0, \end{aligned} \quad (5)$$

where α_i^* determines the desired formation shape, v_c^* determines the desired velocity of the formation, and d_{ij}^* denotes the desired distance between the agent i and j .

3 Main results

In this section, we introduce the agent formation control law and the stability analysis of the system. First, we present the formation control law for triangle configurations. Next, we extend it to the formation control of n agents using angle rigidity. Finally, we discuss a construction scheme for the square formation.

3.1 Triangle formation control

We first consider the desired triangle formation and scaling. The formation control laws for agent #1 to #3 are proposed as follows

$$\begin{aligned} u_1 &= \dot{v}_c^* - k_1 (v_1 - v_c^*) - k_2 (\alpha_1 - \alpha_1^*) (z_{12} + z_{13}), \\ u_2 &= \dot{v}_c^* - k_1 (v_2 - v_c^*) - k_2 (\alpha_2 - \alpha_2^*) (z_{21} + z_{23}), \\ u_3 &= \dot{v}_c^* - k_1 (v_3 - v_c^*) - k_2 (\alpha_3 - \alpha_3^*) (z_{31} + z_{32}) \\ &\quad + k_3 (d_{13} - d_{13}^*) (z_{31} + z_{32}), \end{aligned} \quad (6)$$

where $\dot{v}_c^* \in \mathbb{R}^2$ is the desired acceleration, α_i^* is the desired interior angle of the desired triangle configuration, d_{13}^* is the desired distance between agent #1 and agent #3, and k_1 , k_2 and k_3 are positive scalar gains.

Before presenting the stability analysis, we first provide a Lemma.

Lemma 1 For a matrix M of the form

$$M = \begin{bmatrix} 0_{n \times n} & I_n \\ H_{n \times n} & -kI_n \end{bmatrix}$$

where I is the identity matrix and k is a positive constant. If the submatrix H is Hurwitz and the eigenvalues λ of H satisfy the inequality $(k^2 \operatorname{Re}[\lambda] + \operatorname{Im}^2[\lambda]) < 0$, then the matrix M is also Hurwitz.

Proof. The characteristic polynomial of M can be expressed as

$$\begin{aligned} & |\lambda I_{2n} - M| \\ &= \begin{vmatrix} \lambda I_n & -I_n \\ -H & (\lambda + k)I_n \end{vmatrix} \\ &= |(\lambda + k)I_n| \left| \lambda I_n - \frac{(-I_n)(-H)}{\lambda + k} \right| \\ &= |\lambda(\lambda + k)I_n - H|. \end{aligned} \quad (7)$$

We assume that $\lambda_1, \lambda_2, \dots, \lambda_n$ are the n eigenvalues of H . For any H , there always exists an $n \times n$ unitary matrix U such that

$$H = U \begin{bmatrix} \lambda_1 & * & \cdots & * \\ 0 & \lambda_2 & \cdots & * \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix} U^{-1}. \quad (8)$$

Substituting (8) into $|\lambda(\lambda + k)I_n - H|$, one has

$$\begin{aligned} & |\lambda(\lambda + k)I_n - H| \\ &= (\lambda^2 + k\lambda - \lambda_1)(\lambda^2 + k\lambda - \lambda_2) \cdots (\lambda^2 + k\lambda - \lambda_n). \end{aligned} \quad (9)$$

For any real eigenvalue $\lambda_i, i \in \{1, 2, \dots, n\}$, since H is Hurwitz, λ_i is negative. Therefore, the roots of equation $(\lambda^2 + k\lambda - \lambda_i) = 0$ have negative real parts.

For any pair of complex conjugate eigenvalues, assuming λ_1 and λ_2 are a pair of complex conjugate eigenvalues of H , the polynomial $(\lambda^2 + k\lambda - \lambda_1)(\lambda^2 + k\lambda - \lambda_2)$ can be expressed as follows

$$\begin{aligned} & (\lambda^2 + k\lambda - \lambda_1)(\lambda^2 + k\lambda - \lambda_2) \\ &= \lambda^4 + 2k\lambda^3 + [k^2 - (\lambda_1 + \lambda_2)]\lambda^2 \\ & \quad - k(\lambda_1 + \lambda_2)\lambda + \lambda_1\lambda_2. \end{aligned} \quad (10)$$

Then we have the Hurwitz determinant

$$H_1 = \begin{vmatrix} a_3 & a_1 & 0 & 0 \\ a_4 & a_2 & a_0 & 0 \\ 0 & a_3 & a_1 & 0 \\ 0 & a_4 & a_2 & a_0 \end{vmatrix}, \quad (11)$$

where

$$\begin{aligned} a_4 &= 1, \\ a_3 &= 2k, \\ a_2 &= k^2 - (\lambda_1 + \lambda_2), \\ a_1 &= -k(\lambda_1 + \lambda_2), \\ a_0 &= \lambda_1\lambda_2. \end{aligned} \quad (12)$$

Since $k > 0$ and $\text{Re}[\lambda_j] < 0, j \in \{1, 2\}$, one has

$$\lambda_1 + \lambda_2 = 2 \text{Re}[\lambda_j] < 0, \quad (13)$$

$$\lambda_1\lambda_2 = \text{Re}^2[\lambda_j] + \text{Im}^2[\lambda_j] > 0, \quad (14)$$

where we obtained a_4, a_3, a_2, a_1 and a_0 are all positive. Now, in view of the various orders of leading principal minor determinants of H_1 , we have

$$\Delta_1 = a_3 = 2k > 0, \quad (15)$$

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} a_3 & a_1 \\ a_4 & a_2 \end{vmatrix} \\ &= 2k[k^2 - (\lambda_1 + \lambda_2)] + k(\lambda_1 + \lambda_2) \\ &= 2k^3 - 2k \text{Re}[\lambda_j] > 0, \end{aligned} \quad (16)$$

$$\begin{aligned} \Delta_3 &= a_1\Delta_2 - a_0 \begin{vmatrix} a_3 & a_1 \\ 0 & a_3 \end{vmatrix} \\ &= -2k \text{Re}[\lambda_j](2k^3 - 2k \text{Re}[\lambda_j]) \\ & \quad - 4k^2\lambda_1\lambda_2 \\ &= -4k^2(k^2 \text{Re}[\lambda_j] + \text{Im}^2[\lambda_j]), \end{aligned} \quad (17)$$

$$\Delta_4 = a_0\Delta_3. \quad (18)$$

In view of (17), the roots of the quartic polynomial in (10) have negative real parts if and only if $(k^2 \text{Re}[\lambda_j] + \text{Im}^2[\lambda_j]) < 0, j \in \{1, 2\}$. As a result, the roots of the polynomial (9) have negative real parts if and only if $(k^2 \text{Re}[\lambda_i] + \text{Im}^2[\lambda_i]) < 0, i \in \{1, 2, \dots, n\}$.

The Lemma has been proved.

The Theorem for triangular formation goes as follows.

Theorem 1 Consider a group of three agents with double-integrator dynamics given by (1) under Assumption 1–4. If the initial error $d_{ij}(0) - d_{ij}^*, \alpha_i(0) - \alpha_i^*$ and $v_i(0) - v_i^*$, for $i = 1, 2, 3, j \in \mathcal{N}_i$, are near the equilibrium point, the formation control law (6) ensures that $d_{ij}(t) \rightarrow d_{ij}^*, \alpha_i(t) \rightarrow \alpha_i^*$ and $v_i(t) \rightarrow v_i^*$ asymptotically as $t \rightarrow +\infty$.

Proof. At first, since $\frac{d(\cos \alpha_i)}{dt} = -(\sin \alpha_i) \dot{\alpha}_i$ and from (3), the interior angle dynamics of each agent can be given by

$$\begin{aligned} \dot{\alpha}_1 &= -\frac{z_{13}^T P_{z_{12}}(v_2 - v_1)}{d_{12} \sin \alpha_1} - \frac{z_{12}^T P_{z_{13}}(v_3 - v_1)}{d_{13} \sin \alpha_1}, \\ \dot{\alpha}_2 &= -\frac{z_{23}^T P_{z_{21}}(v_1 - v_2)}{d_{21} \sin \alpha_2} - \frac{z_{21}^T P_{z_{23}}(v_3 - v_2)}{d_{23} \sin \alpha_2}, \\ \dot{\alpha}_3 &= -\frac{z_{32}^T P_{z_{31}}(v_1 - v_3)}{d_{31} \sin \alpha_3} - \frac{z_{31}^T P_{z_{32}}(v_2 - v_3)}{d_{32} \sin \alpha_3}, \end{aligned} \quad (19)$$

where $P_{z_{ij}} = I_2 - z_{ij}z_{ij}^T \in \mathbb{R}^{2 \times 2}$.

Define $e_i = \alpha_i - \alpha_i^*$, then we have the angle error dynamics as follows

$$\dot{e}_i = -N_{jik}(v_k - v_j) - N_{kij}(v_j - v_k), \quad (20)$$

where $N_{jik} = z_{ij}^T \frac{P_{z_{ik}}}{d_{ik} \sin \alpha_{jik}} \in \mathbb{R}^{1 \times 2}, \forall j, i, k \in \{1, 2, 3\}$.

Second, the velocity error dynamics of $v_i - v_c^*$ can be derived from (6), one has

$$\dot{v}_i - \dot{v}_c^* = -k_1(v_i - v_c^*) - k_2(\alpha_i - \alpha_i^*)(z_{i(i-1)} + z_{i(i+1)}). \quad (21)$$

Particularly, since $\alpha_1 + \alpha_2 + \alpha_3 = \pi$, the agent #3's velocity error dynamics can be rewritten as

$$\begin{aligned} \dot{v}_3 - \dot{v}_c^* = & -k_1(v_3 - v_c^*) + k_3(d_{13} - d_{13}^*)(z_{31} + z_{32}) \\ & - k_2(\alpha_1^* - \alpha_1 + \alpha_2^* - \alpha_2)(z_{31} + z_{32}). \end{aligned} \quad (22)$$

Finally, we analyze the dynamic error of relative distance. Since $d_{13}^2 = p_{13}^T p_{13}$, we have

$$2d_{13}\dot{d}_{13} = p_{13}^T \dot{p}_{13} + p_{13}^T \dot{p}_{13}, \quad (23)$$

then

$$\dot{d}_{13} = z_{13}^T [(v_3 - v_c^*) - (v_1 - v_c^*)]. \quad (24)$$

We define the system state variables

$$X = [d_{13} - d_{13}^*, e_1, e_2, v_1^T - v_c^{*T}, v_2^T - v_c^{*T}, v_3^T - v_c^{*T}]^T, \quad (25)$$

where $X \in \mathbb{R}^9$. According to (19)-(24), we have the closed loop system as follows

$$\dot{X} = AX = \begin{bmatrix} 0_{3 \times 3} & R_1 \\ R_2 & -k_1 I_6 \end{bmatrix} X, \quad (26)$$

where $R_1 \in \mathbb{R}^{3 \times 6}, R_2 \in \mathbb{R}^{6 \times 3}$, and

$$R_1 = \begin{bmatrix} -z_{13}^T & 0 & z_{13}^T \\ N_{312} + N_{213} & -N_{312} & -N_{213} \\ -N_{321} & N_{32} + N_{123} & -N_{123} \end{bmatrix}, \quad (27)$$

$$R_2 = \begin{bmatrix} 0 & -k_2(z_{12} + z_{13}) & 0 \\ 0 & 0 & -k_2(z_{21} + z_{23}) \\ k_3(z_{31} + z_{32}) & k_2(z_{31} + z_{32}) & k_2(z_{31} + z_{32}) \end{bmatrix}. \quad (28)$$

Then, we linearize R_1 and R_2 near the equilibrium $X = 0$ to analyze the stability of the system, one has

$$\dot{X} = \bar{A}X = \begin{bmatrix} 0_{3 \times 3} & \bar{R}_1 \\ \bar{R}_2 & -k_1 I_6 \end{bmatrix} X, \quad (29)$$

where $\bar{A}$ is the linearization of the system matrix near the origin. For convenience, the symbol $(\bar{\cdot})$ denotes that the variable

is assigned a value at $X = 0$ below. Next, we prove that system (29) is stable by analyzing the eigenvalues of $\bar{A}$, and prove that $\bar{A}$ is Hurwitz. Let λ be an eigenvalue of $\bar{A}$, then we have characteristic equation as follows

$$\begin{aligned} & |\lambda I_9 - \bar{A}| \\ &= \begin{vmatrix} \lambda I_3 & -\bar{R}_1 \\ -\bar{R}_2 & (\lambda + k_1) I_6 \end{vmatrix} \\ &= (\lambda + k_1)^3 |\lambda(\lambda + k_1) I_3 - \bar{R}_1 \bar{R}_2|. \end{aligned} \quad (30)$$

From (30), we find that $-k_1$ is the triple root of $\bar{A}$, also, $-k_1 < 0$. Then, we analyze the eigenvalues of matrix $\bar{R}_1 \bar{R}_2$. Let $\bar{G}_1 = \bar{R}_1 \bar{R}_2$ one has

$$\bar{G}_1 = \begin{bmatrix} \bar{G}_{11} & \bar{G}_{12} & \bar{G}_{13} \end{bmatrix}, \quad (31)$$

where $\bar{G}_{1i} \in \mathbb{R}^{3 \times 1}$ is the i th column of $\bar{G}_1, i \in \{1, 2, 3\}$. By using $P_{z_{ij}} z_{ij} = 0$, we have $N_{jik} z_{ik} = 0$ and $N_{jik} z_{ki} = 0$. Then we can simplify to obtain

$$\bar{G}_{11} = \begin{bmatrix} k_3 z_{13}^T (z_{31} + z_{32}) \\ -k_3 N_{213} z_{32} \\ -k_3 N_{123} z_{31} \end{bmatrix}, \quad (32)$$

$$\bar{G}_{12} = \begin{bmatrix} k_2 z_{13}^T (z_{12} + z_{32}) \\ -k_2 (N_{312} z_{13} + N_{213} z_{12} + N_{213} z_{32}) \\ k_2 (N_{321} z_{13} - N_{123} z_{31}) \end{bmatrix}, \quad (33)$$

$$\bar{G}_{13} = \begin{bmatrix} k_2 z_{13}^T (z_{31} + z_{32}) \\ k_2 (N_{312} z_{23} - N_{213} z_{32}) \\ -k_2 (N_{321} z_{23} + N_{123} z_{21} + N_{123} z_{31}) \end{bmatrix}. \quad (34)$$

Simplify further by using $\cos \alpha_{jik} = z_{ij}^T z_{ik}$, one has

$$\bar{G}_1 = \begin{bmatrix} -k_3(1 + \cos \alpha_3) & -k_2(\cos \alpha_3 - \cos \alpha_1) & -k_2(1 + \cos \alpha_3) \\ -k_3 f_{13} & -k_2(g_1 + f_{13}) & -k_2(f_{13} - f_{12}) \\ -k_3 f_{23} & -k_2(f_{23} - f_{21}) & -k_2(g_2 + f_{23}) \end{bmatrix}, \quad (35)$$

where $f_{ij} = \frac{\sin \alpha_j}{d_{ij}}$ and $g_i = (\sin \alpha_i) \left(\frac{1}{d_{ij}} + \frac{1}{d_{ik}} \right), i, j, k \in \{1, 2, 3\}$, and $j, k \in \mathcal{N}_i$.

Let $p = 1 + \cos \alpha_3$ and $q = \cos \alpha_3 - \cos \alpha_1$, (35) can be rewritten as

$$\bar{G}_1 = \begin{bmatrix} -k_3 p & -k_2 q & -k_2 p \\ -k_3 f_{13} & -k_2(g_1 + f_{13}) & -k_2(f_{13} - f_{12}) \\ -k_3 f_{23} & -k_2(f_{23} - f_{21}) & -k_2(g_2 + f_{23}) \end{bmatrix}. \quad (36)$$

Then, let λ_1 be an eigenvalue of $\bar{G}_1$, one has

$$\begin{aligned} & |\lambda_1 I - \bar{G}_1| \\ &= \begin{vmatrix} \lambda_1 + k_3 p & k_2 q & k_2 p \\ k_3 f_{13} & \lambda_1 + k_2 (g_1 + f_{13}) & k_2 (f_{13} - f_{12}) \\ k_3 f_{23} & k_2 (f_{23} - f_{21}) & \lambda_1 + k_2 (g_2 + f_{23}) \end{vmatrix} \\ &= a_3 \lambda_1^3 + a_2 \lambda_1^2 + a_1 \lambda_1 + a_0, \end{aligned} \quad (37)$$

where

$$a_3 = 1,$$

$$a_2 = k_2 f_{13} + k_2 f_{23} + k_2 g_1 + k_2 g_2 + k_3 p,$$

$$\begin{aligned} a_1 &= k_2^2 f_{13} f_{21} + k_2^2 f_{12} f_{23} + k_2^2 f_{13} g_2 \\ &\quad + k_2^2 f_{23} g_1 + k_2^2 (g_1 g_2 - f_{12} f_{21}) \\ &\quad + k_2 k_3 f_{13} (p - q) + k_2 k_3 g_1 p + k_2 k_3 g_2 p, \end{aligned}$$

$$a_0 = k_2^2 k_3 [f_{12} f_{23} (p - q) + f_{13} g_2 (p - q) + (g_1 g_2 - f_{12} f_{21}) p]. \quad (38)$$

Now, we focus on the coefficients of the roots of the cubic polynomial of (37) by using the condition of Routh-Hurwitz criteria. Considering $p > 0$, $p - q > 0$ and $g_1 g_2 - f_{12} f_{21} > 0$, we can get that a_3 , a_2 , a_1 and a_0 are all positive. Then, we obtained the Hurwitz determinant

$$H = \begin{vmatrix} a_2 & a_3 & 0 \\ a_0 & a_1 & a_2 \\ 0 & 0 & a_0 \end{vmatrix}, \quad (39)$$

one has

$$\Delta_1 = a_2 > 0, \quad (40)$$

$$\Delta_2 = \begin{vmatrix} a_2 & a_3 \\ a_0 & a_1 \end{vmatrix} = a_2 a_1 - a_3 a_0 > 0, \quad (41)$$

$$\Delta_3 = a_0 \Delta_2 > 0. \quad (42)$$

where inequality (41) is calculated in Appendix. By checking the above Routh-Hurwitz conditions, we obtained the characteristic roots of Eq.(37) all have negative real parts. Therefore, $\bar{G}_1$ is Hurwitz.

In summary, $\bar{A}$ is Hurwitz under the Lemma, and it can be concluded that the agent formation system is exponentially stable when the initial errors are located near the equilibrium point.

Remark 1 It should be pointed out that, for the proposed control law, the number of distance measurements is reduced. When it is necessary to control expansion and contraction to avoid obstacles, it is not necessary for each agent to measure distance information, nor is it necessary for a large number of agents to measure distance information.

Remark 2 In view of Fig. 4, note that the local bearings of each agent can be misaligned. Even if the agents lose their global sense of direction in certain complex environments, it will not affect the completion of the formation. The local bearing misalignment does not affect the interior angle α_i and the distance d_{ij} measurements. Regardless of the bearing orientation, the agents' formation motion is always toward the angular bisector $z_{ij} + z_{ik}$ of the interior angle α_i , and the direction of $z_{ij} + z_{ik}$ is constant.

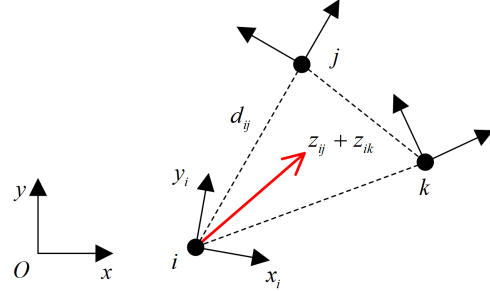


Figure 4 The local bearings are not aligned.

3.2 Polygonal formation

In this subsection, we will give the polygonal formation control law of n agents. We consider a formation of n agents and expand the team scale of the formation through the method shown in Fig. 5. For agent i , $i > 3$, each agent needs to measure two angles $\alpha_{j_1 i j_2}^*$ and $\alpha_{j_2 i j_3}^*$ relative to three neighbor agents $j_1, j_2, j_3 < i$, $j_1, j_2, j_3 \in \mathcal{N}_i$. We have an angle set $\mathcal{A} = \{(j_1, i, j_2), (j_2, i, j_3)\}$.

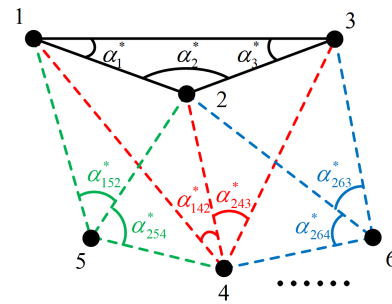


Figure 5 Expansion of formation team scale.

Assumption 5 When using the above method to expand the team scale of the formation, we assume that the desired positions of agent $i > 3$, agent j_1 , agent j_2 and agent j_3 are not in a circle, $\alpha_{j_1 i j_2}^*$ and $\alpha_{j_2 i j_3}^*$ are not zero or π .

Remark 3 When Assumption 5 holds and the formation configuration is constructed in the manner depicted in Fig. 5, the formation is angularly rigid, and the analysis of angle

rigidity can be found in [26].

To solve the problem proposed in Section II, the formation control laws are designed as

$$u_i = \dot{v}_c^* - k_1 (v_i - v_c^*) + k_3 (d_{13} - d_{13}^*) (z_{ij} + z_{ik}) - k_2 \sum_{(j,i,k) \in \mathcal{A}} (\alpha_{jik} - \alpha_{jik}^*) (z_{ij} + z_{ik}), \quad (43)$$

where k_1 and k_2 are positive scalar gains, k_3 is a positive constant if $i = 3$, and $k_3 = 0$ otherwise.

Theorem 2 Consider a group of n agents with double-integrator dynamics given by (1) under Assumption 1 – 5. If the initial error $d_{ij}(0) - d_{ij}^*$, $\alpha_i(0) - \alpha_i^*$ and $v_i(0) - v_c^*$ are near the equilibrium point, the formation control law (43) ensures that $d_{ij}(t) \rightarrow d_{ij}^*$, $\alpha_i(t) \rightarrow \alpha_i^*$ and $v_i(t) \rightarrow v_c^*$ asymptotically as $t \rightarrow +\infty$.

Proof. The system state variables are defined as

$$X = [d_{13} - d_{13}^*, e_1, e_2, e_{41}, e_{42}, \dots, e_{n1}, e_{n2}, v_1^T - v_c^{*T}, v_2^T - v_c^{*T}, v_3^T - v_c^{*T}, \dots, v_n^T - v_c^{*T}]^T \in \mathbb{R}^{4n-3}. \quad (44)$$

According to the previous foreshadowing, we directly give the closed-loop dynamics

$$\dot{X} = BX = \begin{bmatrix} 0_{(2n-3) \times (2n-3)} & S_1 \\ S_2 & -k_1 I_{2n} \end{bmatrix} X, \quad (45)$$

where $S_1 \in \mathbb{R}^{(2n-3) \times 2n}$, $S_2 \in \mathbb{R}^{2n \times (2n-3)}$, and

$$S_1 = \begin{bmatrix} -z_{13}^T & 0 & z_{13}^T & 0 & \dots & 0 \\ N_{312} + N_{213} & -N_{312} & -N_{213} & 0 & \dots & 0 \\ -N_{321} & N_{32} + N_{123} & -N_{123} & 0 & \dots & 0 \\ -N_{241} & -N_{142} & 0 & N_{241} + N_{142} & \dots & 0 \\ 0 & -N_{342} & -N_{243} & N_{342} + N_{243} & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots & \dots \end{bmatrix}, \quad (46)$$

$$S_2 = \begin{bmatrix} 0 & -k_2(z_{12} + z_{13}) & 0 & \dots & 0 \\ 0 & 0 & -k_2(z_{21} + z_{33}) & \dots & 0 \\ k_3(z_{31} + z_{32}) & k_2(z_{31} + z_{32}) & k_2(z_{31} + z_{32}) & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \end{bmatrix}. \quad (47)$$

Then, we linearize S_1 and S_2 near the equilibrium $X = 0$ to analyze the stability of the system, one has

$$\dot{X} = \bar{B}X = \begin{bmatrix} 0_{(2n-3) \times (2n-3)} & \bar{S}_1 \\ \bar{S}_2 & -k_1 I_{2n} \end{bmatrix} X. \quad (48)$$

We have the characteristic equation as follows

$$\begin{aligned} & |\lambda I_{4n-3} - \bar{B}| \\ &= \begin{vmatrix} \lambda I_{2n-3} & -\bar{S}_1 \\ -\bar{S}_2 & (\lambda + k_1) I_{2n} \end{vmatrix} \\ &= (\lambda + k_1)^3 |\lambda(\lambda + k_1) I_{2n-3} - \bar{S}_1 \bar{S}_2|, \end{aligned} \quad (49)$$

and in (49)

$$\bar{S}_1 \bar{S}_2 = \begin{bmatrix} \bar{G}_1 & 0 & 0 & \dots & 0 \\ * & \bar{G}_4 & 0 & \dots & 0 \\ * & * & \bar{G}_5 & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ * & * & * & \dots & \bar{G}_n \end{bmatrix}, \quad (50)$$

where $\bar{G}_1 \in \mathbb{R}^{3 \times 3}$, $\bar{G}_4, \bar{G}_5, \dots, \bar{G}_n \in \mathbb{R}^{2 \times 2}$, and matrix $\bar{S}_1 \bar{S}_2$ is a block lower triangular matrix. For $\bar{G}_i$, $i > 3$

$$\bar{G}_i = k_2 \bar{F}_i = k_2 \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \quad (51)$$

where

$$\begin{aligned} a &= -k_2 (N_{j_1 i j_2} + N_{j_2 i j_1}) (z_{i j_1} + z_{i j_2}), \\ b &= -k_2 (N_{j_1 i j_2} + N_{j_2 i j_1}) (z_{i j_2} + z_{i j_3}), \\ c &= -k_2 (N_{j_2 i j_3} + N_{j_3 i j_2}) (z_{i j_1} + z_{i j_2}), \\ d &= -k_2 (N_{j_2 i j_3} + N_{j_3 i j_2}) (z_{i j_2} + z_{i j_3}). \end{aligned}$$

Let λ_1 and λ_2 be the two eigenvalues of $\bar{F}_i$ respectively. By using

$$\begin{cases} \lambda_1 + \lambda_2 = \text{tr}(\bar{F}_i) < 0, \\ \lambda_1 \lambda_2 = \det(\bar{F}_i) > 0, \end{cases} \quad (52)$$

we have $\bar{G}_i$ is Hurwitz. Then, substituting (50) into (49), the characteristic polynomial can be rewritten as

$$\begin{aligned} & |\lambda I_{4n-3} - \bar{B}| \\ &= (\lambda + k_1)^3 |\lambda(\lambda + k_1) I_3 - \bar{G}_1| \prod_{i=4}^n |\lambda(\lambda + k_1) I_2 - \bar{G}_i|. \end{aligned} \quad (53)$$

Since $\bar{G}_i$ is Hurwitz, we give the conclusion directly that Theorem 2 is proved under the Lemma.

Remark 4 It should be pointed out that there is angle rigidity under Assumption 5. However, the formation scale is not rigid. Below we explain why only the change in distance between a pair of agents can determine the scale of the entire formation.

As shown in Fig. 2, according to the law of sines

$$\frac{d_{23}}{\sin \alpha_1^*} = \frac{d_{13}}{\sin \alpha_2^*} = \frac{d_{12}}{\sin \alpha_3^*} = 2R, \quad (54)$$

where R denotes the radius of the circumcircle of $\triangle 123$, when d_{13} shrinks $k = \frac{d_{13}}{d_{13}^*}$ times, the radius of the circumcircle also shrinks k times.

Then, since a chord and an angle can uniquely determine a circumcircle, when the angle is determined, the radius of the circumscribed circle is only related to the length of the chord. For example, the chord 12 and α_{142}^* is given in Fig. 6(a) and Fig. 6(b). As a result, the scale of a formation of four agents can be completely determined by the distance on either side of the first triangle $\triangle 123$.

According to mathematical induction, it can be derived to n agents.

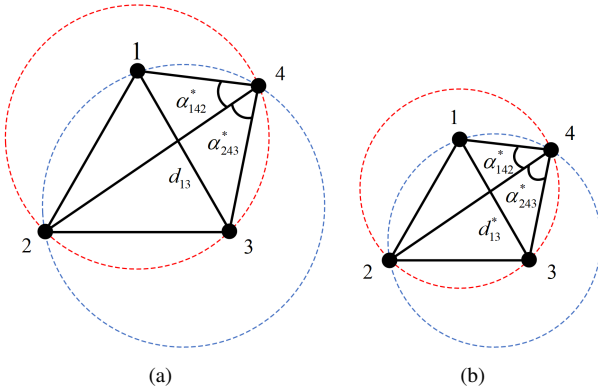


Figure 6 The uniqueness of a formation.

3.3 Square formation control

In this subsection, we discuss the control for square formation with four agents on a circle.

As shown in Fig. 6(a), the desired formations of agent #4 and the first three agents are not on the same circle, which satisfies Assumption 5. This constitutes a condition for the validity of the angle rigidity theory. Therefore, the desired position of agent #4 is unique under the constraints of α_{142}^* and α_{243}^* . Here, we relax the condition. When the four agents are on a circle, that is, when the red circle and the blue circle coincide in Fig. 6, the formation configuration satisfies the expected angle and distance, however, the configuration is not unique as shown in Fig. 7. Agent #4 measures the LOS angle to agent #1, #2 and #3 respectively, forming angle constraints, where $\alpha_{142}^* = \pi/4$ and $\alpha_{243}^* = \pi/4$ in Fig. 7(a). Obviously, the position of agent #4 is not unique, because agent #4 can move arbitrarily along the arc 143 as shown in Fig. 7(b).

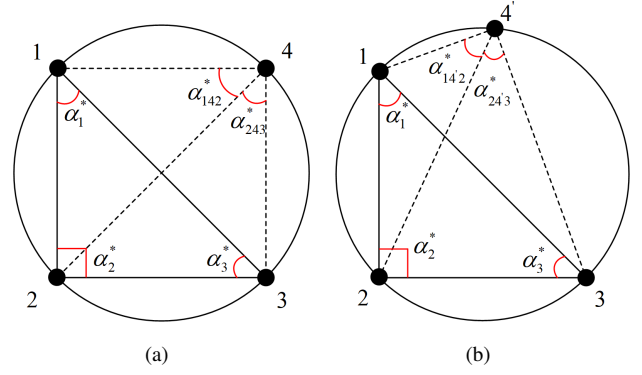
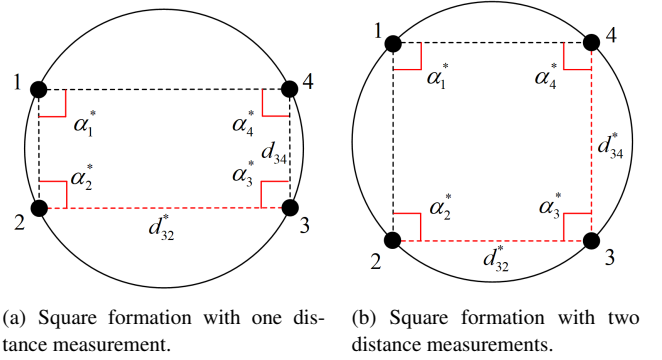


Figure 7 Uncertainty of formation on a circle.



(a) Square formation with one distance measurement. (b) Square formation with two distance measurements.

Figure 8 Square formation construction.

In the configuration depicted in Fig. 8(a), where agent #3 measures only d_{32} , d_{34} may be ambiguous. To solve the above problems, we can change the constraints as shown in Fig. 8(b). agent #1, agent #2, agent #3 and agent #4 each measure the interior angle α_1 , α_2 , α_3 and α_4 with its neighbors. Then agent #3 measures the distance d_{32} and d_{34} . Therefore, the problem can be formulated as follows

$$\begin{aligned} \lim_{t \rightarrow +\infty} e_i &= \lim_{t \rightarrow +\infty} (\alpha_i - \pi/2) = 0, \\ \lim_{t \rightarrow +\infty} (d_{32} - d_{32}^*) &= 0, \\ \lim_{t \rightarrow +\infty} (d_{34} - d_{34}^*) &= 0. \end{aligned} \quad (55)$$

The square formation control laws for agent #1 to #4 are proposed to be

$$\begin{aligned} u_i &= \dot{v}_c^* - k_1 (v_i - v_c^*) - k_2 (\alpha_i - \alpha_i^*) (z_{i(i-1)} + z_{i(i+1)}), i \in \{1, 2, 4\}, \\ u_3 &= \dot{v}_c^* - k_1 (v_3 - v_c^*) - k_2 (\alpha_3 - \alpha_3^*) (z_{32} + z_{34}) \\ &\quad + k_3 (d_{32} - d_{32}^*) z_{32} + k_3 (d_{34} - d_{34}^*) z_{34}. \end{aligned} \quad (56)$$

Theorem 3 Consider a group of four agents with double-integrator dynamics given by (1) under Assumption 1 – 4.

$$\begin{bmatrix} \bar{T}_1 \\ \bar{T}_2 \end{bmatrix} \begin{bmatrix} \bar{T}_3 & \bar{T}_4 \end{bmatrix} = \begin{bmatrix} -k_3 & -k_3 \cos \alpha_3 & 0 & k_2(1 + \cos \alpha_2) & k_2(1 + \cos \alpha_3) \\ -k_3 \cos \alpha_3 & -k_3 & -k_2(1 + \cos \alpha_4) & -k_2(1 + \cos \alpha_4) & k_2(\cos \alpha_3 - \cos \alpha_4) \\ 0 & 0 & -k_2 \left(\frac{\sin \alpha_1}{d_{12}} + \frac{\sin \alpha_1}{d_{14}} + \frac{\sin \alpha_4}{d_{14}} \right) & k_2 \left(\frac{\sin \alpha_2}{d_{12}} - \frac{\sin \alpha_4}{d_{14}} \right) & -k_2 \frac{\sin \alpha_4}{d_{14}} \\ 0 & -k_3 \frac{\sin \alpha_3}{d_{23}} & k_2 \frac{\sin \alpha_1}{d_{12}} & -k_2 \left(\frac{\sin \alpha_2}{d_{23}} + \frac{\sin \alpha_2}{d_{12}} \right) & k_2 \frac{\sin \alpha_3}{d_{23}} \\ k_3 \frac{\sin \alpha_3}{d_{34}} & k_3 \frac{\sin \alpha_3}{d_{23}} & -k_2 \frac{\sin \alpha_4}{d_{34}} & k_2 \left(\frac{\sin \alpha_2}{d_{23}} - \frac{\sin \alpha_4}{d_{34}} \right) & -k_2 \left(\frac{\sin \alpha_3}{d_{34}} + \frac{\sin \alpha_3}{d_{23}} + \frac{\sin \alpha_4}{d_{34}} \right) \end{bmatrix} \quad (57)$$

If the initial error $d_{ij}(0) - d_{ij}^*, \alpha_i(0) - \alpha_i^*$ and $v_i(0) - v_c^*$, for $i = 1, 2, 3, 4$, $j \in \mathcal{N}_i$, are near the equilibrium point, the square formation control law (56) ensures that $d_{ij}(t) \rightarrow d_{ij}^*$, $\alpha_i(t) \rightarrow \alpha_i^*$ and $v_i(t) \rightarrow v_c^*$ asymptotically as $t \rightarrow +\infty$.

Proof. With the control law (56), the angle error dynamics of each agent can be given by

$$\dot{e}_i = -N_{jik}(v_k - v_j) - N_{kij}(v_j - v_k). \quad (58)$$

The velocity error dynamics are given by

$$\begin{aligned} \dot{v}_i - v_c^* &= -k_1(v_i - v_c^*) - k_2(\alpha_i - \alpha_i^*)(z_{i(i-1)} + z_{i(i+1)}), i \in \{1, 2, 4\}, \\ \dot{v}_3 - v_c^* &= -k_1(v_3 - v_c^*) - k_2(\alpha_3 - \alpha_3^*)(z_{32} + z_{34}) \\ &\quad + k_3(d_{32} - d_{32}^*)z_{32} + k_3(d_{34} - d_{34}^*)z_{34}. \end{aligned} \quad (59)$$

The distance error dynamics are given by

$$\begin{aligned} \dot{d}_{32} &= z_{32}^T [(v_2 - v_c^*) - (v_3 - v_c^*)], \\ \dot{d}_{34} &= z_{34}^T [(v_4 - v_c^*) - (v_3 - v_c^*)]. \end{aligned} \quad (60)$$

Since $\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 = 2\pi$, let the system state variables be

$$X = [d_{32} - d_{32}^*, d_{34} - d_{34}^*, e_1, e_2, e_3, v_1^T - v_c^{*T}, v_2^T - v_c^{*T}, v_3^T - v_c^{*T}, v_4^T - v_c^{*T}]^T \in \mathbb{R}^{13}. \quad (61)$$

Then we have the closed-loop system linearized at the equilibrium point as follows

$$\dot{X} = \bar{C}X = \begin{bmatrix} 0_{2 \times 2} & 0_{2 \times 3} & \bar{T}_1 \\ 0_{3 \times 2} & 0_{3 \times 3} & \bar{T}_2 \\ \bar{T}_3 & \bar{T}_4 & -k_1 I_8 \end{bmatrix} X, \quad (62)$$

where $\bar{T}_1 \in \mathbb{R}^{2 \times 8}$, $\bar{T}_2 \in \mathbb{R}^{3 \times 8}$, $\bar{T}_3 \in \mathbb{R}^{8 \times 2}$, $\bar{T}_4 \in \mathbb{R}^{8 \times 3}$, and

$$\begin{aligned} \bar{T}_1 &= \begin{bmatrix} 0 & z_{32}^T & -z_{32}^T & 0 \\ 0 & 0 & -z_{34}^T & z_{34}^T \end{bmatrix}, \\ \bar{T}_2 &= \begin{bmatrix} N_{412} + N_{214} & -N_{412} & 0 & -N_{214} \\ -N_{321} & N_{123} + N_{321} & -N_{123} & 0 \\ 0 & -N_{432} & N_{234} + N_{432} & -N_{234} \end{bmatrix}, \\ \bar{T}_3 &= \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ k_3 z_{32} & k_3 z_{34} \\ 0 & 0 \end{bmatrix}, \\ \bar{T}_4 &= \begin{bmatrix} -k_2(z_{12} + z_{14}) & 0 & 0 \\ 0 & -k_2(z_{21} + z_{23}) & 0 \\ 0 & 0 & -k_2(z_{32} + z_{34}) \\ k_2(z_{41} + z_{43}) & k_2(z_{41} + z_{43}) & k_2(z_{41} + z_{43}) \end{bmatrix}. \end{aligned} \quad (63)$$

Thus, we can obtain the characteristic equation

$$\begin{aligned} &|\lambda I_{13} - \bar{C}| \\ &= \begin{vmatrix} \lambda I_5 & -\begin{bmatrix} \bar{T}_1 \\ \bar{T}_2 \end{bmatrix} \\ -\begin{bmatrix} \bar{T}_3 & \bar{T}_4 \end{bmatrix} & (\lambda + k_1) I_8 \end{vmatrix} \\ &= (\lambda + k_1)^3 \left| \lambda (\lambda + k_1) I_5 - \begin{bmatrix} \bar{T}_1 \\ \bar{T}_2 \end{bmatrix} \begin{bmatrix} \bar{T}_3 & \bar{T}_4 \end{bmatrix} \right|, \end{aligned} \quad (64)$$

where $\begin{bmatrix} \bar{T}_1 & \bar{T}_2 \end{bmatrix}^T \begin{bmatrix} \bar{T}_3 & \bar{T}_4 \end{bmatrix}$ is given by (57). Let $H = \begin{bmatrix} \bar{T}_1 & \bar{T}_2 \end{bmatrix}^T \begin{bmatrix} \bar{T}_3 & \bar{T}_4 \end{bmatrix}$. The determinant $|\lambda(\lambda + k_1) I_5 - H|$ is equivalent to the characteristic equation associated with the matrix $\begin{bmatrix} 0 & I_5 \\ H & -k_1 I_5 \end{bmatrix}$. Thus, from the lemma, if H is Hurwitz, we have $\bar{C}$ is Hurwitz so that the agent square formation system is exponentially stable.

Remark 5 Note that the movement direction of the convergence part of the distance error is changed in control law (56). Previously, in the control law (6), the movement direction of the dynamic error of the distance was the angle bisector $z_{ij} + z_{ik}$, that is, the weight of the dynamic error of the distance in z_{ij} and z_{ik} was equal. Here, the weights on z_{ij} and z_{ik} become the respective distance dynamic errors. See in Fig. 9, the direction of movement becomes $e_{d_{ij}z_{ij}} + e_{d_{ik}z_{ik}}$, where $e_{d_{ij}} = d_{ij} - d_{ij}^*$.

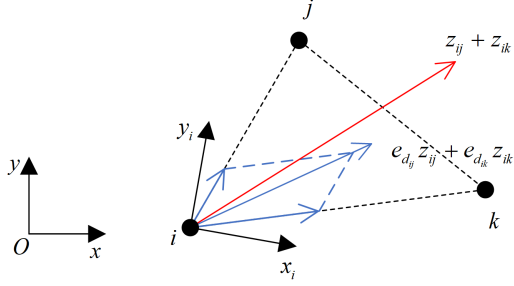


Figure 9 Direction of motion caused by range dynamic error.

4 Simulations

In this section, we use the simplified model of the quadrotor UAV to verify the validity of the theoretical results in Section III. The low-level control of each quadrotor is determined by an outer position loop and an inner attitude loop [31]. The position model can be decoupled into two linear models, namely the horizontal channel model and the altitude channel model [32, Chapter 11]:

$$\begin{cases} \dot{p}_h = v_h, \\ \dot{v}_h = -gA_\psi\Theta_h, \end{cases} \quad (65)$$

$$\begin{cases} \dot{p}_z = v_z, \\ \dot{v}_z = g - \frac{f}{m}, \end{cases} \quad (66)$$

where

$$p_h = \begin{bmatrix} p_x \\ p_y \end{bmatrix}, R_\psi = \begin{bmatrix} \cos \psi & -\sin \psi \\ \sin \psi & \cos \psi \end{bmatrix}, \quad (67)$$

$$A_\psi = R_\psi \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}, \Theta_h = \begin{bmatrix} \phi \\ \theta \end{bmatrix}.$$

Here, $g \in \mathbb{R}$ is the gravitational constant, $f \in \mathbb{R}$ is the total thrust, $m \in \mathbb{R}$ is the mass of the QUAV, and $\phi \in (-\pi/2, \pi/2)$, $\theta \in (-\pi/2, \pi/2)$, and $\psi \in (-\pi, \pi)$ denote the roll, pitch, and yaw angles, respectively.

Therefore, we present the control structure as shown in Fig. 10. The position loop in the horizontal channel is controlled by the formation control law proposed in this paper, while the attitude loop is controlled by a PD controller. Assume that all quadrotors move within the X-Y plane, with their altitude and yaw angle remaining constant, i.e., the altitude channel input $u_{zd} = 0$ and the yaw angle $\psi_d = 0$, where $(\cdot)_d$ represents the desired value. The input of the horizontal channel $u_{hd} \in \mathbb{R}^2$ is updated at each time step according to (6), (43), and (56). Then, ϕ_d and θ_d can be obtained by inverting (65). Ultimately, the rotational speeds $\omega_{d,k}$ of the four rotors are obtained, where $k = 1, 2, 3, 4$.

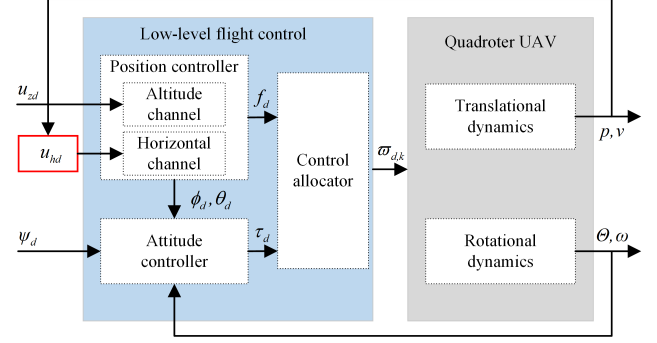


Figure 10 The control structure of the QUAVs.

4.1 Triangular formation maneuvering case

We first consider the triangular formation maneuvering case. The initial formation of the three quadrotors is an isosceles right triangle, where the initial position coordinates and velocity are selected as $p_1(0) = [0, 0]^T m$, $p_2(0) = [4, 0]^T m$, $p_3(0) = [0, 4]^T m$, $v_1(0) = [0, 0]^T m/s$, $v_2(0) = [0, 0]^T m/s$, $v_3(0) = [0, 0]^T m/s$. Then, let the desired formation configuration be the equilateral triangle, i.e. $\alpha_1^* = \alpha_2^* = \alpha_3^* = \pi/3$.

Applying the control law (6), the simulation results of the formation maneuvering are shown in Fig. 11. The control gains are selected as $k_1 = 0.2$, $k_2 = 0.1$, $k_3 = 0.02$. The desired translational velocity is selected as $v_c^* = [0.5, 0]^T m/s$. And the desired distance d_{13}^* is given by

$$d_{13}^*(t) = \begin{cases} 10 \text{ m}, & 0 < t \leq 120 \text{ s}, \\ 5 \text{ m}, & 120 \text{ s} < t \leq 240 \text{ s}, \\ 10 \text{ m}, & t > 240 \text{ s}. \end{cases} \quad (68)$$

In Fig. 11, the dotted line depicts the movement trajectories of the quadrotors. First, it shows the evolution from an initial formation to an equilateral triangle formation of the desired scale. Then, changing the desired distance d_{13}^* when the formation flying encounters narrow terrain allows the formation to maintain an equilateral triangle formation through the terrain. Finally, the scale of the formation is restored after passing the terrain. Fig. 13 shows the evolution trajectories of angle error, distance error, and actual distances during the entire formation flight. The simulation results show the stability of the control law (6).

4.2 Polygonal formation maneuvering case

In this case, we consider the desired formation configuration of six quadrotors. The initial configuration of the formation is given arbitrarily. The initial position coordinates and velocity are selected as $p_1(0) = [0, 0]^T m$, $p_2(0) = [2, 2]^T m$, $p_3(0) = [-1, 4]^T m$, $p_4(0) = [5, 3]^T m$, $p_5(0) = [4, 0]^T m$, $p_6(0) =$

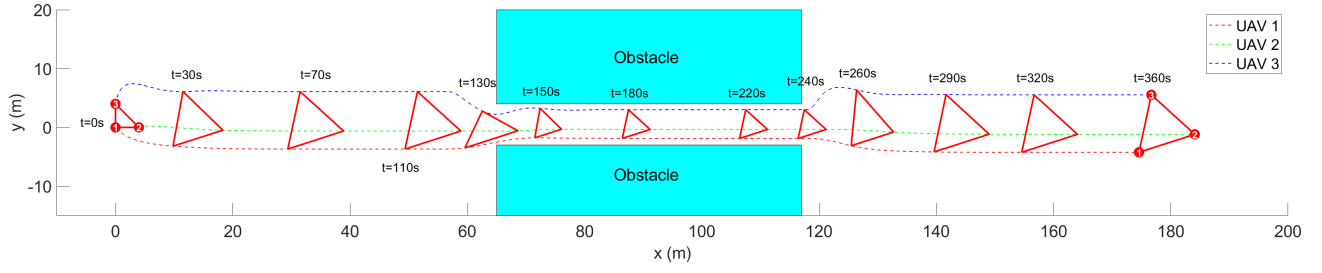


Figure 11 The triangle formation maneuvering trajectories when encountering narrow terrain.

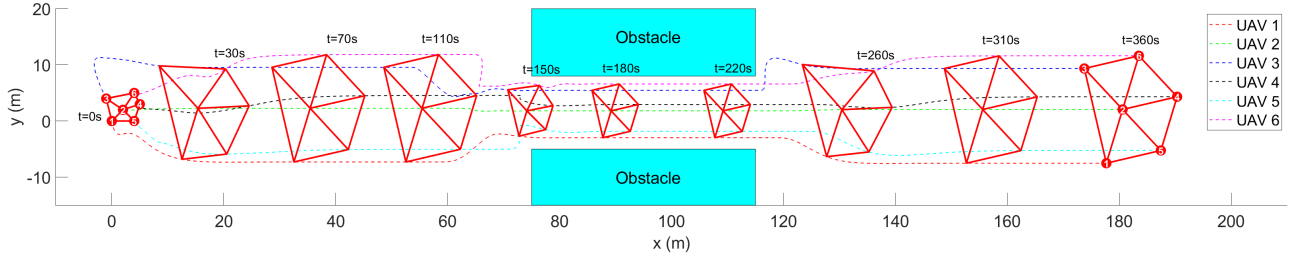


Figure 12 The polygonal formation maneuvering trajectories when encountering narrow terrain.

$[4, 5]^T m$, $v_1(0) = [0, 0]^T m/s$, $v_2(0) = [0, 0]^T m/s$, $v_3(0) = [0, 0]^T m/s$, $v_4(0) = [0, 0]^T m/s$, $v_5(0) = [0, 0]^T m/s$, $v_6(0) = [0, 0]^T m/s$. Then, let the desired formation configuration be the square, i.e. $\alpha_{213}^* = \pi/6$, $\alpha_{123}^* = 2\pi/3$, $\alpha_{132}^* = \pi/6$, $\alpha_{142}^* = \pi/6$, $\alpha_{243}^* = \pi/6$, $\alpha_{152}^* = \pi/3$, $\alpha_{254}^* = \pi/3$, $\alpha_{263}^* = \pi/3$, $\alpha_{264}^* = \pi/3$.

Applying the control law (43), the simulation results of the formation maneuvering are shown in Fig. 12. The control gains are selected as $k_1 = 0.3$, $k_2 = 0.4$, $k_3 = 0.03$. The desired translational velocity is selected as $v_c^* = [0.5, 0]^T m/s$. The desired distance d_{13}^* is given by (68).

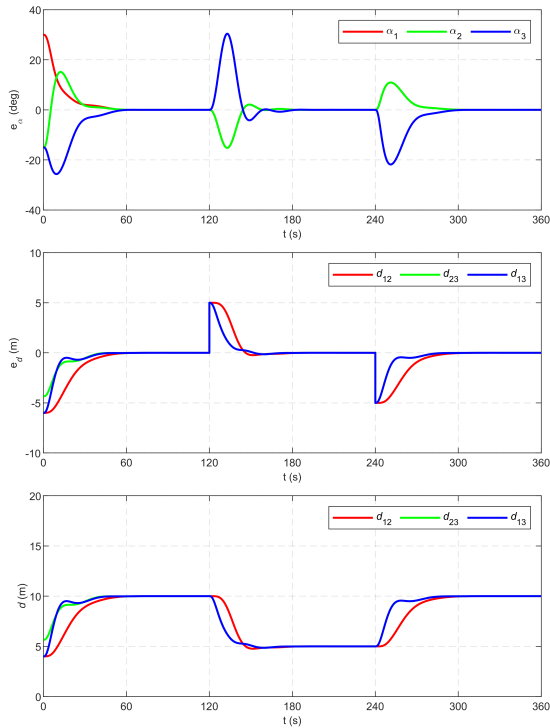


Figure 13 Angle and distance error trajectories and actual distance evolution trajectories for the triangular formation case.

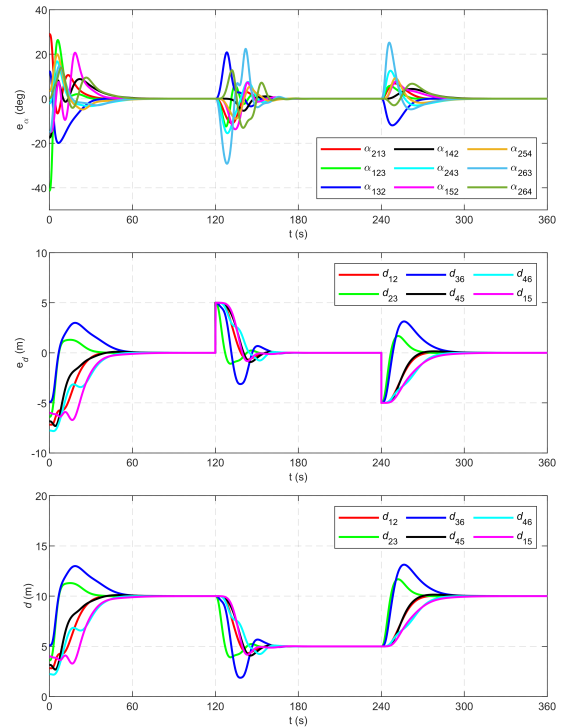


Figure 14 Angle and distance error trajectories and actual distance evolution trajectories for the polygonal formation case.

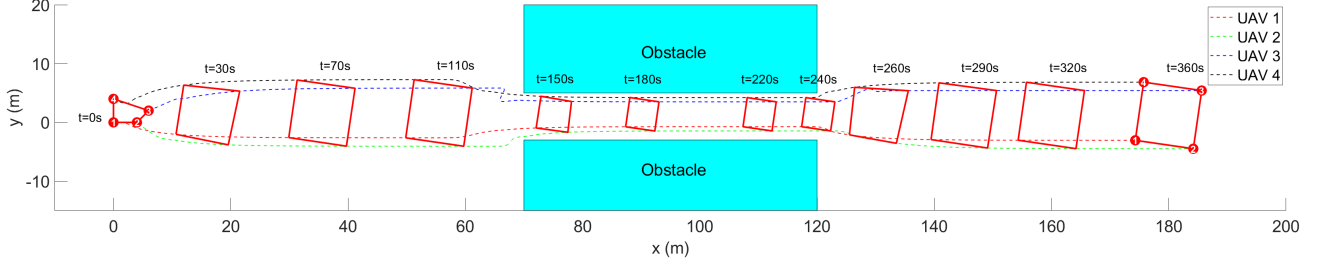


Figure 15 The square formation maneuvering trajectories when encountering narrow terrain.

First, Fig. 12 shows the evolution from the initial formation to a polygonal formation of the desired size. Then, when the formation encounters narrow terrain during flight, by changing the desired distance d_{13}^* , the formation is able to maintain its original geometric topology while passing through the terrain. Finally, the formation restores its size after passing through the terrain. Fig. 14 shows the evolution trajectories of angle error, distance error, and actual distances during the entire formation flight. The simulation results show the stability of the control law (43).

4.3 Square formation maneuvering control on a circle case

In this case, we consider the desired formation configuration of four quadrotors on a circle. The initial configuration of the formation is given arbitrarily. The initial position coordinates and velocity are selected as $p_1(0) = [0, 0]^T m$, $p_2(0) = [4, 0]^T m$, $p_3(0) = [6, 2]^T m$, $p_4(0) = [0, 4]^T m$, $v_1(0) = [0, 0]^T m/s$, $v_2(0) = [0, 0]^T m/s$, $v_3(0) = [0, 0]^T m/s$, $v_4(0) = [0, 0]^T m/s$. Then, let the desired formation configuration be the square, i.e. $\alpha_1^* = \alpha_2^* = \alpha_3^* = \alpha_4^* = \pi/2$.

Applying the control law (56), the simulation results of the formation maneuvering are shown in Fig. 15. The control gains are selected as $k_1 = 0.5$, $k_2 = 0.5$, $k_3 = 0.1$. The desired translational velocity is selected as $v_c^* = [0.5, 0]^T m/s$. The desired distance d_{32}^* and d_{34}^* are given by

$$d_{32}^*(t) = d_{34}^*(t) = \begin{cases} 10 \text{ m}, & 0 < t \leq 120 \text{ s}, \\ 5 \text{ m}, & 120 \text{ s} < t \leq 240 \text{ s}, \\ 10 \text{ m}, & t > 240 \text{ s}. \end{cases} \quad (69)$$

First, Fig. 15 shows the evolution from the initial formation to a square formation of the desired size. Then, when the formation encounters narrow terrain during flight, by changing the desired distances d_{32}^* and d_{34}^* , the formation is able to maintain its original square shape while passing through the terrain. Finally, the formation restores its size after passing through the terrain. Fig. 16 shows the evolution trajectories of angle error, distance error, and actual distances during the entire formation flight. The simulation results show the stability of the control law (56).

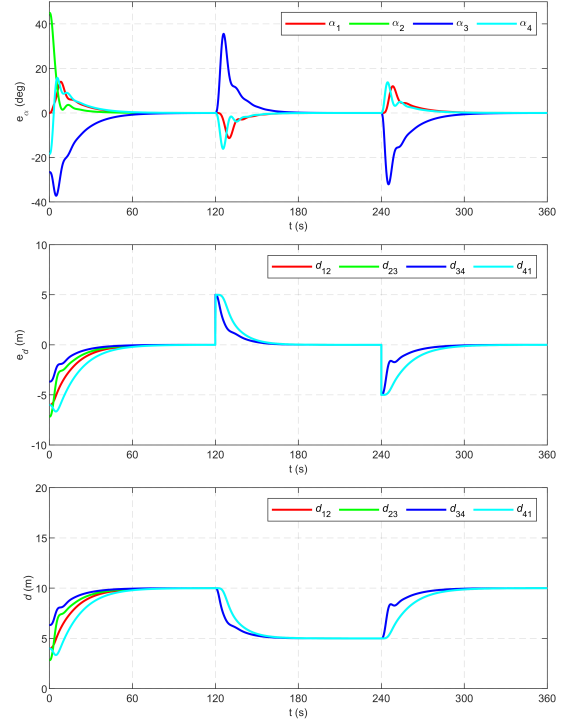


Figure 16 Angle and distance error trajectories and actual distance evolution trajectories for the square formation case.

It can be seen from Fig. 11, Fig. 12, and Fig. 15 that the direction of the entire formation relative to the global coordinate system after deformation and expansion is uncertain, precisely because we are using local information. However, in all, the purpose of scaling the deformation and formation has been achieved. In addition, when the formation encounters obstacles that require scaling transformations, we only consider maintaining the size of the bounding box to ensure safe passage through narrow spaces.

5 Conclusion and discussion

This paper investigates formation control problems in multi-agent systems and designs control laws that regulate deformation and formation scaling using angle and distance information. Each agent measures the angle, with only one agent

measuring the distance. This approach effectively addresses the ambiguity in controlling the scale of the formation when relying solely on angle information. The overall scaling of the formation is controlled by a single agent's distance measurement. When a rigid formation scales to avoid obstacles, it does not require extensive adjustments of distance constraints between different agents. Multiple experiments have been conducted to demonstrate the effectiveness of the proposed control algorithms. The results show that agent formations can navigate through narrow terrains with flexible sizes.

Currently, we can only prove the local convergence of the closed-loop nonlinear formation double integrator dynamics; global convergence remains unknown. Although the method is feasible for n agents, its performance and complexity may be affected as the number of agents increases significantly. Due to the diverse geometric shapes of formations with different numbers of agents, discussing all possible configurations in detail is impractical. For obstacle avoidance tasks, the primary goal is to avoid obstacles, and adopting a feasible formation scheme is sufficient without considering arbitrary formation shapes. Future work will address global convergence issues.

6 Appendix

For Section III-A, see inequality (41), substituting a_3 , a_2 , a_1 and a_0 into inequality (41) and organizing, we have

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} a_2 & a_3 \\ a_0 & a_1 \end{vmatrix} \\ &= (k_2^3 f_{23} + k_2^2 k_3 p) g_1^2 + 2k_2^2 k_3 p g_1 g_2 \\ &\quad + k_2^2 k_3 g_1 [f_{13}(p - q) + 2f_{23}p + f_{13}p] + k_2^2 k_3^2 p^2 g_1 \\ &\quad + (k_2^3 f_{13} + k_2^2 k_3 p) g_2^2 + k_2^2 k_3 f_{13}^2 (p - q) \\ &\quad + k_2^2 k_3 (2f_{13}p + f_{23}p) g_2 + k_2^2 k_3^2 p^2 g_2 + k_2^2 k_3 f_{13} f_{21} p \\ &\quad + 2k_2^3 (f_{13} + f_{23}) g_1 g_2 + k_2^3 g_1^2 g_2 + k_2^3 g_1 g_2^2 \quad (70) \\ &\quad + k_2^3 g_1 (f_{13} f_{21} - f_{12} f_{21} + f_{12} f_{23} + f_{13} f_{23} + f_{23}^2) \\ &\quad + k_2^3 g_2 (f_{13} f_{21} - f_{12} f_{21} + f_{12} f_{23} + f_{13} f_{23} + f_{13}^2) \\ &\quad + k_2^3 (f_{13} + f_{23}) (f_{13} f_{21} - f_{12} f_{21} + f_{12} f_{23}) \\ &\quad + k_2^2 k_3 f_{13} f_{23} p - k_2^2 k_3 f_{13} f_{23} q + k_2^2 k_3^2 f_{13} p (p - q) \\ &\quad + k_2^2 k_3 f_{12} f_{23} q, \end{aligned}$$

where for each term with a negative sign in (70), a positive term can be found and the result of adding it is positive by using $p > 0$, $p - q > 0$ and $g_1 g_2 - f_{12} f_{21} > 0$, except for the last term whose sign is uncertain.

Now, let us analyze the positive or negative of the last term. Considering $k_2^2 k_3 f_{12} f_{23} q$, there are the following cases that determine the positivity of (70).

Case 1 If

$$d_{13} < d_{12} < d_{23}, \quad (71)$$

or

$$d_{12} < d_{13} < d_{23}, \quad (72)$$

or

$$d_{12} < d_{23} < d_{13}, \quad (73)$$

we have $\alpha_3 < \alpha_1$ and $\cos \alpha_3 > \cos \alpha_1$, as a result, $q > 0$, then $\Delta_2 > 0$.

Case 2 If

$$d_{13} < d_{23} < d_{12}, \quad (74)$$

or

$$d_{23} < d_{13} < d_{12}, \quad (75)$$

we have $\alpha_3 > \alpha_1$ and $\cos \alpha_3 < \cos \alpha_1$, then $q < 0$. But by using the law of sines (54), one has

$$\begin{aligned} f_{13} - f_{12} &= \frac{\sin \alpha_3}{d_{13}} - \frac{\sin \alpha_2}{d_{12}} = 2R \frac{d_{12}^2 - d_{13}^2}{d_{12} d_{13}} > 0 \\ -f_{13} f_{23} q &> -f_{12} f_{23} q \\ -f_{13} f_{23} q + f_{12} f_{23} q &> 0 \\ -k_2^2 k_3 f_{13} f_{23} q + k_2^2 k_3 f_{12} f_{23} q &> 0, \end{aligned} \quad (76)$$

so that $\Delta_2 > 0$.

Case 3 If

$$d_{23} < d_{12} < d_{13}, \quad (77)$$

we have

$$f_{32} - f_{12} = \frac{\sin \alpha_2}{d_{23}} - \frac{\sin \alpha_2}{d_{12}} > 0. \quad (78)$$

There are two cases: when α_1 , α_2 and α_3 are all acute angles, one has

$$p + q = 1 + \cos \alpha_3 + \cos \alpha_3 - \cos \alpha_1 > 0. \quad (79)$$

And when α_2 is an obtuse angle, α_1 and α_3 are acute angles, there is also

$$p + q > 0, \quad (80)$$

where we obtained

$$\begin{aligned} f_{12} f_{23} p + f_{12} f_{23} q &> 0, \\ f_{32} f_{23} p + f_{12} f_{23} q &> 0. \end{aligned} \quad (81)$$

By using $g_2 = f_{32} + f_{12}$, one has

$$k_2^2 k_3 f_{32} f_{23} p + k_2^2 k_3 f_{12} f_{23} q > 0, \quad (82)$$

where we have $\Delta_2 > 0$.

In conclusion, it has been proven that $\Delta_2 > 0$ is valid.

This work was supported in part by the National Natural Science Foundation of China (Grant No. 12272116 and No. 62088101), by Zhejiang Provincial Natural Science Foundation of China (Grant No. LY22A020007), and by the Fundamental Research Funds for the Provincial Universities of Zhejiang GK239909299001-014.

Conflict of interest The authors declare that they have no conflict of interest.

- 1 Cai H, Su Y, Huang J. Cooperative control of multi-agent systems[J]. Cham: Springer-Verlag, 2022
- 2 Liu Y, Liu J, He Z, et al. A survey of multi-agent systems on distributed formation control[J]. *Unmanned Systems*, 2024, 12(05): 913-926
- 3 Alotaibi E T, Alqefari S S, Koubaa A. Lsar: Multi-uav collaboration for search and rescue missions[J]. *IEEE Access*, 2019, 7: 55817-55832
- 4 Yufka A, Parlaktuna O, Ozkan M. Formation-based cooperative transportation by a group of non-holonomic mobile robots[C]//2010 IEEE International Conference on Systems, Man and Cybernetics. IEEE, 2010: 3300-3307
- 5 Liu H, Tian Y, Lewis F L. Robust trajectory tracking in satellite time-varying formation flying[J]. *IEEE Transactions on Cybernetics*, 2020, 51(12): 5752-5760
- 6 Oh K K, Park M C, Ahn H S. A survey of multi-agent formation control[J]. *Automatica*, 2015, 53: 424-440
- 7 Li B, Zhang J, Dai L, et al. A hybrid offline optimization method for reconfiguration of multi-UAV formations[J]. *IEEE Transactions on Aerospace and Electronic Systems*, 2020, 57(1): 506-520
- 8 Wang Y, Yue Y, Shan M, et al. Formation reconstruction and trajectory replanning for multi-UAV patrol[J]. *IEEE/ASME Transactions on Mechatronics*, 2021, 26(2): 719-729
- 9 Tron R, Thomas J, Loianno G, et al. A distributed optimization framework for localization and formation control: Applications to vision-based measurements[J]. *IEEE Control Systems Magazine*, 2016, 36(4): 22-44
- 10 Liu G, Li B, Ji Y. A modified HP-adaptive pseudospectral method for multi-UAV formation reconfiguration[J]. *ISA transactions*, 2022, 129: 217-229
- 11 Ze K, Wang W, Liu K, et al. Time-Varying Formation Planning and Distributed Control for Multiple UAVs in Clutter Environment[J]. *IEEE Transactions on Industrial Electronics*, 2023
- 12 Lee B H, Ahn H S. Distributed formation control via global orientation estimation[J]. *Automatica*, 2016, 73: 125-129
- 13 Meng Z, Anderson B D O, Hirche S. Formation control with mismatched compasses[J]. *Automatica*, 2016, 69: 232-241
- 14 de Marina H G. Maneuvering and robustness issues in undirected displacement-consensus-based formation control[J]. *IEEE Transactions on Automatic Control*, 2020, 66(7): 3370-3377
- 15 Zhao S, Zelazo D. Bearing rigidity and almost global bearing-only formation stabilization[J]. *IEEE Transactions on Automatic Control*, 2015, 61(5): 1255-1268
- 16 Hendrickson B. Conditions for unique graph realizations[J]. *SIAM journal on computing*, 1992, 21(1): 65-84
- 17 Asimow L, Roth B. The rigidity of graphs, II[J]. *Journal of Mathematical Analysis and Applications*, 1979, 68(1): 171-190
- 18 Oh K K, Ahn H S. Leader-follower type distance-based formation control of a group of autonomous agents[J]. *International Journal of Control, Automation and Systems*, 2017, 15: 1738-1745
- 19 Zhao S, Zelazo D. Bearing-based formation stabilization with directed interaction topologies[C]//2015 54th IEEE Conference on Decision and Control (CDC). IEEE, 2015: 6115-6120
- 20 Zhao S, Li Z, Ding Z. Bearing-only formation tracking control of multiagent systems[J]. *IEEE Transactions on Automatic Control*, 2019, 64(11): 4541-4554
- 21 Zhao S, Zelazo D. Translational and scaling formation maneuver control via a bearing-based approach[J]. *IEEE Transactions on Control of Network Systems*, 2015, 4(3): 429-438
- 22 Hu R, Huang X, Xu C. Integrated visual navigation based on angles-only measurements for asteroid final landing phase[J]. *Astrodynamics*, 2023, 7(1): 69-82
- 23 Gong B, Wang S, Li S, et al. Review of space relative navigation based on angles-only measurements[J]. *Astrodynamics*, 2023, 7(2): 131-152
- 24 Jing G, Zhang G, Lee H W J, et al. Angle-based shape determination theory of planar graphs with application to formation stabilization[J]. *Automatica*, 2019, 105: 117-129
- 25 Li J, Chen L, Cao M, et al. Satellite formation flying control by using only angle measurements[J]. *IEEE Transactions on Aerospace and Electronic Systems*, 2022, 59(2): 1439-1451
- 26 Chen L, Cao M, Li C. Angle rigidity and its usage to stabilize multi-agent formations in 2-D[J]. *IEEE Transactions on Automatic Control*, 2020, 66(8): 3667-3681
- 27 Chen L, Shi M, de Marina H G, et al. Stabilizing and maneuvering angle rigid multiagent formations with double-integrator agent dynamics[J]. *IEEE Transactions on Control of Network Systems*, 2022, 9(3): 1362-1374
- 28 Liu T, de Queiroz M. Distance+ angle-based control of 2-d rigid formations[J]. *IEEE transactions on cybernetics*, 2020, 51(12): 5969-5978
- 29 Shao Z, Zhou Z, Wang Y, et al. Directed Formation Control Using Distance and Angle in Local Coordinate Frameworks[J]. *IEEE Transactions on Industrial Electronics*, 2024
- 30 Zhao S. Affine formation maneuver control of multiagent systems[J]. *IEEE Transactions on Automatic Control*, 2018, 63(12): 4140-4155
- 31 Dong X, Yu B, Shi Z, et al. Time-varying formation control for unmanned aerial vehicles: Theories and applications[J]. *IEEE Transactions on Control Systems Technology*, 2014, 23(1): 340-348
- 32 Quan Q. Introduction to multicopter design and control[M]. Singapore: Springer, 2017